Circle: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Circle geometry facts

Formulas

Chord: $2r\sin\theta$ (angle 2θ at centre), distance $r\cos\theta$
Chords / secants: $PA \cdot PB = PC \cdot PD = PT^2$

KEY POINTS

- Equal chords $\iff$ equal central angles $\iff$ equidistant from centre; nearer chord is longer
- Angle at centre $= 2\times$ angle at circumference; same segment $\Rightarrow$ equal angles; semicircle $\Rightarrow 90°$
- Cyclic quadrilateral: opposite angles supplementary; alternate segment theorem
- Tangents from a point are equal; CP bisects their angle and is the perpendicular bisector of the chord of contact

2. Equation of a circle

Formulas

$(x-h)^2 + (y-k)^2 = r^2$; $x^2 + y^2 + 2gx + 2fy + c = 0$
Centre $(-g, -f)$, $r = \sqrt{g^2 + f^2 - c}$

KEY POINTS

- 2nd degree equation is a circle iff coeff x^2 = coeff y^2 and no xy term
- $g^2 + f^2 - c > 0$ real, $= 0$ point circle, < 0 imaginary
- Three points: solve for g, f, c; four points concyclic: fourth satisfies the circle of three
- Image in a line: same radius, centre reflected

Watch out

Divide by the coefficient of x^2 before reading g, f, c

3. Diametric form and intercepts

Formulas

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

Chord $2\sqrt{r^2 - p^2}$; x-int $2\sqrt{g^2 - c}$, y-int $2\sqrt{f^2 - c}$

KEY POINTS

- Smallest circle through A, B: AB as diameter
- Circle on a chord as diameter: add the x- and y-quadratics of the end points (equal leading coefficients)
- x-axis: $g^2 > c$ cuts, $g^2 = c$ touches, $g^2 < c$ misses (same with f for y-axis)
- Touching x-axis: $|k| = r$; both axes: $(x \mp r)^2 + (y \mp r)^2 = r^2$
- Any curve: chord $= |x_1 - x_2|\sqrt{1 + m^2}$

4. Parametric form and point position

Formulas

Point: $(h + r\cos\theta,\ k + r\sin\theta)$
Distance of A: max $AC + r$, min $|AC - r|$

KEY POINTS

- $S_1 > 0$ outside, $= 0$ on, < 0 inside
- T: $x^2 \to xx_1$, $2x \to x + x_1$, $2xy \to xy_1 + yx_1$
- Max of $px + qy$ on a circle: $ph + qk + r\sqrt{p^2 + q^2}$
- Nearest / farthest points lie on line AC; line to circle: $p - r$
- Region questions: combine $S_1 < 0$ with the side of a line

5. Line and circle

Formulas

Tangency $p = r$; $y = mx + c$ touches $x^2 + y^2 = a^2$ iff $c^2 = a^2(1 + m^2)$

KEY POINTS

- $p > r$ misses, $p = r$ tangent, $p < r$ secant, $p = 0$ diameter
- Any curve: substitute and use D of the quadratic
- Circle touching two lines: centre on a bisector, r = distance
- Line between two circles: distance from each centre > its radius, centres on opposite sides

6. Tangent and normal

Formulas

At (x_1, y_1): $T = 0$; parametric $x\cos\theta + y\sin\theta = a$; slope $y = mx \pm a\sqrt{1 + m^2}$
Chord $P(\alpha)Q(\beta)$: $x\cos\frac{\alpha+\beta}{2} + y\sin\frac{\alpha+\beta}{2} = a\cos\frac{\alpha-\beta}{2}$

STEPS

1. Point of tangency given: $T = 0$
2. Other condition: assume the line, apply $p = r$
3. From an external point: $y - y_1 = m(x - x_1)$; a lost root means $x = x_1$

KEY POINTS

- Every normal passes through the centre
- Point of contact = foot of perpendicular from the centre

7. Tangent lengths and power

Formulas

$L = \sqrt{S_1}$; power $= S_1 = PA \cdot PB$

$$PACB = RL,\ AB = \frac{2RL}{\sqrt{R^2 + L^2}},\ \triangle PAB = \frac{RL^3}{R^2 + L^2}$$

KEY POINTS

- Angle between tangents 2θ: $\tan\theta = \frac{R}{L}$, $\sin\theta = \frac{R}{CP}$
- Circumcircle of PAB: CP as diameter, $(x - x_1)(x + g) + (y - y_1)(y + f) = 0$
- Power: + outside, 0 on, − inside

8. Director circle

Formulas

$x^2 + y^2 = 2a^2$ (radius $\sqrt{2}\,r$, concentric)

KEY POINTS

- Locus of the meet of perpendicular tangents
- Constant angle 2θ between tangents: $CP = r\,\mathrm{cosec}\,\theta$
- Any curve: slope-form tangent through (h, k), put $m_1 m_2 = -1$

9. Chord with given mid-point

Formulas

$T = S_1$; length $2\sqrt{r^2 - CM^2}$

KEY POINTS

- Mid-point of a given chord: foot of perpendicular from the centre
- Shortest chord through an interior M: bisected at M
- Chords through fixed P: mid-points on the circle on CP as diameter
- Chords subtending 2θ at the centre: mid-points on a concentric circle of radius $r\cos\theta$
- Chord subtending 90° at the origin: homogenise the curve with $T = S_1$

10. Chord of contact, pair of tangents, pole and polar

Formulas

Chord of contact / polar: $T = 0$; pair of tangents: $SS_1 = T^2$
Pole of $lx + my = n$ w.r.t. $x^2 + y^2 = a^2$: $\left(\frac{a^2 l}{n}, \frac{a^2 m}{n}\right)$

KEY POINTS

- Tangents at the ends of a chord meet at its pole: compare $T = 0$ with the chord
- Points on a fixed line: chords of contact pass through the pole of the line
- Tangents at α, β meet at $\left(\frac{a\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, \frac{a\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}\right)$
- Conjugate points: each on the other's polar; harmonic conjugate of P lies on its polar

Watch out

Point inside: polar exists but no chord of contact

11. Family of circles

Formulas

Two circles: $S_1 + \lambda S_2 = 0$; circle and line: $S + \lambda L = 0$
Through A, B: (circle on AB) $+\lambda$ (line AB)
Touching L at (x_1, y_1): $(x - x_1)^2 + (y - y_1)^2 + \lambda L = 0$
Triangle: $L_1L_2 + \lambda L_2L_3 + \mu L_3L_1 = 0$; quadrilateral: $L_1L_3 + \lambda L_2L_4 = 0$

KEY POINTS

- $S + \lambda L = 0$ always passes through the two fixed points $S = L = 0$
- Touching a circle at P: $S + \lambda T = 0$
- Circumcircle from lines: coeff x^2 = coeff y^2, coeff $xy = 0$

Watch out

No solution for λ: the answer may be $S_2 = 0$ itself ($\lambda \to \infty$)

12. Common tangents

Formulas

$L_{DCT} = \sqrt{d^2 - (r_1 - r_2)^2}$, $L_{TCT} = \sqrt{d^2 - (r_1 + r_2)^2}$
Angle: $\sin\theta = \frac{r_1 \mp r_2}{d}$ (half-angle; $-$ DCT, $+$ TCT)

KEY POINTS

- DCTs meet dividing C_1C_2 externally in $r_1 : r_2$; TCTs internally
- Centres on the same side of a DCT, opposite sides of a TCT
- Equal radii: DCTs parallel to C_1C_2
- Touching circles: common tangent at contact $S_1 - S_2 = 0$

STEPS

1. Find P or R by section formula
2. Draw tangents from it to one circle ($p = r$)

13. Two circles: relative position

Formulas

$d > r_1 + r_2$: 4; $d = r_1 + r_2$: 3; $|r_1 - r_2| < d < r_1 + r_2$: 2; $d = |r_1 - r_2|$: 1; $d < |r_1 - r_2|$: 0 common tangents

KEY POINTS

- Least distance: separate $d - r_1 - r_2$, inside $|r_1 - r_2| - d$, else 0
- Point of contact divides C_1C_2 in $r_1 : r_2$
- Circle touching a fixed circle and a fixed line: centre on a parabola

14. Radical axis, radical centre, coaxial circles

Formulas

Radical axis / common chord: $S_1 - S_2 = 0$
Coaxial family: $S_1 + \lambda(S_1 - S_2) = 0$

KEY POINTS

- Perpendicular to C_1C_2; bisects every common tangent; none for concentric circles
- Intersecting: common chord; touching: common tangent at contact
- S_2 bisects the circumference of S_1: C_1 lies on $S_1 - S_2 = 0$
- Radical centre: equal tangents to three circles; for circles on the sides of a triangle as diameters it is the orthocentre

Watch out

Make the coefficients of x^2, y^2 equal to 1 in both circles before subtracting

15. Angles and orthogonality

Formulas

Line and circle: $\cos\theta = \frac{p}{r}$; two circles: $\cos\theta = \dfrac{r_1^2 + r_2^2 - d^2}{2r_1r_2}$
Orthogonal: $r_1^2 + r_2^2 = d^2 \iff 2g_1g_2 + 2f_1f_2 = c_1 + c_2$

KEY POINTS

- Angle not defined if no common point; tangent line: angle 0
- Line cuts a circle orthogonally $\iff$ it passes through the centre
- Orthogonal to three circles: centre = radical centre, r = tangent length
- Centres of circles orthogonal to two circles: on their radical axis
- Orthogonal circles: common chord $= \dfrac{2r_1r_2}{d}$

16. Common mistakes

STEPS

1. Reading g, f, c before making the coefficient of x^2 equal to 1
2. Forgetting the vertical tangent $x = x_1$ from an external point
3. Taking only one sign in $|k| = r$, $p = r$ (several circles possible)
4. Using $SS_1 = T^2$ or $T = 0$ for a point inside the circle as a chord of contact
5. Mixing $r_1 - r_2$ (DCT) and $r_1 + r_2$ (TCT)
6. Missing $\lambda \to \infty$ in $S_1 + \lambda S_2 = 0$
7. Internal vs external touching: $d = r_1 + r_2$ vs $d = |r_1 - r_2|$
8. Strict vs non-strict inequalities in 'inside' and 'does not meet' conditions